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DEPARTMENT OF ARTIFICIAL TECHNOLOGY AND MACHINE LEARNING
CCW - AMT 305 INTRODUCTION TO MACHINE LEARNING MCQs

MODULE 1

1.Which of the following is NOT a machine learning paradigm?
a) Supervised learning
b) Reinforcement learning
c) Algorithmic learning
d) Unsupervised learning

2. Which machine learning paradigm uses labeled data for training?
a) Supervised learning
b) Semi-supervised learning
c) Unsupervised learning
d) Reinforcement learning

3. In which learning paradigm does the model learn from rewards and punishments?
a) Supervised learning
b) Unsupervised learning
c) Reinforcement learning
d) Semi-supervised learning

4. Which of the following is an example of unsupervised learning?
a) Image classification
b) Spam detection
c) Clustering
d) Sentiment analysis

5. Which learning method is a combination of supervised and unsupervised learning?
a) Semi-supervised learning
b) Reinforcement learning
c) Deep learning
d) Active learning

6. Which of the following best describes a hypothesis class in machine learning?
a) A set of possible models that can be chosen for training
b) A collection of labeled data

c) A method for evaluating performance
d) A technique for feature engineering

7. What is the function of a version space in machine learning?
a) It defines the error of a model
b) It represents the set of hypotheses consistent with the training data
c) It determines the best algorithm for learning
d) It optimizes the model's accuracy

8. What does the VC Dimension measure?
a) The complexity of a learning algorithm
b) The maximum number of samples that a classifier can perfectly classify
c) The number of hidden layers in a neural network
d) The bias in a learning model

9. In Probably Approximately Correct (PAC) learning, what does "approximately correct" mean?
a) The model is always accurate
b) The model has a high probability of being within an acceptable error margin
c) The model is guaranteed to be error-free
d) The model is trained with 100% labeled data

10. Which of the following is a major challenge in supervised learning?
a) Lack of labeled data
b) Lack of computational power
c) Too many hidden layers
d) Redundant outputs

11. What is the impact of noise in a training dataset?
a) Improves model accuracy
b) Helps in better feature selection
c) Increases error rate and decreases performance
d) Has no effect on the model

12. Which of the following can help mitigate the impact of noise in supervised learning?
a) Adding more noise
b) Feature selection and data preprocessing
c) Ignoring the noisy data completely
d) Removing all low-value features

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13. What is the primary challenge in multi-class classification?
a) Selecting the right optimizer
b) Handling multiple outputs and decision boundaries
c) Managing a single label for classification
d) Using linear regression instead of classification

14. Which algorithm is commonly used for multi-class classification?
a) Logistic Regression
b) k-Nearest Neighbors
c) Support Vector Machines
d) All of the above

15. One-vs-All and One-vs-One are techniques used for:
a) Dimensionality reduction
b) Handling missing values
c) Multi-class classification
d) Model evaluation

16.What is the main goal of model selection?
a) Choosing the best dataset
b) Selecting the model that generalizes well to new data
c) Maximizing the training accuracy
d) Reducing the number of features

17. What problem occurs when a model performs well on training data but poorly on test data?
a) Underfitting
b) Overfitting
c) Data leakage
d) Noise removal

18. What technique helps to reduce overfitting?
a) Increasing the training data size
b) Adding more complex features
c) Using a model with a higher VC dimension
d) Removing regularization

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19. Which of the following is NOT a technique to improve generalization?
a) Cross-validation
b) Regularization
c) Data augmentation
d) Increasing model complexity

20. What does cross-validation help with in model training?
a) Increases training accuracy
b) Reduces bias and improves model generalization
c) Selects the most complex model
d) Ensures 100% accuracy

MODULE 2

21. Which of the following is a dimensionality reduction technique?
a) Decision Trees
b) Principal Component Analysis (PCA)
c) Logistic Regression
d) Naïve Bayes

22. What is the primary purpose of dimensionality reduction?
a) Increase the number of features in a dataset
b) Improve model complexity
c) Reduce computational cost and avoid overfitting
d) Add more noise to the data

23. Which of the following is NOT a method for dimensionality reduction?
a) Subset Selection
b) PCA
c) Random Forest
d) Feature Extraction

24. What does Principal Component Analysis (PCA) aim to do?
a) Find the most important features in a dataset
b) Predict categorical values
c) Perform supervised learning
d) Reduce the number of dependent variables

25. Which of the following is NOT true about PCA?
a) It reduces the dimensions of the dataset
b) It maintains most of the variance in the data
c) It increases the number of variables in a dataset
d) It projects data onto new feature axes

26. Which of the following is a type of regression?
a) Linear Regression
b) k-Nearest Neighbors
c) Decision Tree
d) Naïve Bayes

27. Which of the following methods is used to find the optimal parameters in linear regression?
a) Gradient Descent
b) Clustering
c) Decision Trees
d) Naïve Bayes

28. Which method is commonly used to solve linear regression equations?
a) k-Means Clustering
b) Matrix Method
c) Naïve Bayes
d) Principal Component Analysis

29. What is the main assumption of linear regression?
a) The relationship between variables is non-linear
b) The relationship between independent and dependent variables is linear
c) The dataset has a large number of missing values
d) The data is categorical

30. Which of the following is an indication of overfitting in regression?
a) High training accuracy but low test accuracy
b) High test accuracy but low training accuracy
c) High accuracy on both training and test data
d) Low variance in predictions

31. What is the objective of the gradient descent algorithm?
a) Maximize the error function

b) Minimize the loss function
c) Reduce bias completely
d) Increase the number of features

32. Which of the following is a type of gradient descent?
a) Batch Gradient Descent
b) Feature Gradient Descent
c) Component Gradient Descent
d) PCA Gradient Descent

33. What is the major drawback of gradient descent?
a) It does not work for linear regression
b) It may get stuck in local minima
c) It requires a large amount of labeled data
d) It is always computationally inexpensive

34. Which of the following is NOT a linear classification method?
a) Logistic Regression
b) Naïve Bayes
c) Decision Tree (ID3)
d) Support Vector Machine (SVM)

35. Which classification method is based on Bayes Theorem?
a) Logistic Regression
b) Naïve Bayes
c) Decision Tree
d) k-Nearest Neighbors

36. What is the key assumption of Naïve Bayes classification?
a) Features are independent of each other
b) Features are highly correlated
c) It only works for large datasets
d) It does not use probability for classification

37. Which classification algorithm is commonly used for text classification problems?
a) Naïve Bayes
b) Principal Component Analysis
c) k-Means Clustering
d) Decision Trees

38. What does logistic regression predict?
a) Continuous values
b) Categorical outcomes
c) Clusters in data
d) Principal components

39. Which of the following is a decision tree algorithm used for classification?
a) ID3
b) k-Means
c) PCA
d) Gradient Descent

40. What does the ID3 algorithm use to split data at each node?
a) Gradient Descent
b) Entropy and Information Gain
c) Principal Components
d) Naïve Bayes Assumptions

MODULE 3

41. Which metric measures the proportion of correctly classified positive samples?
a) Precision
b) Recall
c) Accuracy
d) F1-Score

42. What does Recall measure in classification?
a) Ratio of correctly predicted positive observations to all actual positives
b) Ratio of correctly predicted negative observations to all actual negatives
c) Ratio of true positives to true negatives
d) Ratio of false positives to true positives

43. What is the harmonic mean of Precision and Recall?
a) Accuracy
b) ROC Curve
c) F1-Score
d) AUC

44. What does the Receiver Operating Characteristic (ROC) curve represent?
a) The relationship between precision and recall
b) The trade-off between true positive rate and false positive rate
c) The difference between training and testing accuracy
d) The accuracy of a classification model

45. What does AUC (Area Under Curve) measure in an ROC curve?
a) The number of false positives
b) The probability of a model distinguishing between classes
c) The difference between accuracy and recall
d) The overall number of misclassification

46. Which metric is most useful when dealing with imbalanced datasets?
a) Accuracy
b) F1-Score
c) ROC-AUC
d) Precision

47. Which of the following is NOT a classification performance measure?
a) Recall
b) Precision
c) Gradient Descent
d) Accuracy

48. What is Cross Validation used for in machine learning?
a) Reducing the size of datasets
b) Evaluating model performance on different training and testing sets
c) Increasing the learning rate of a model
d) Finding the number of features in a dataset

49. What is the purpose of Bootstrapping in machine learning?
a) Selecting the best activation function
b) Creating multiple training samples from existing data
c) Increasing the number of features
d) Reducing bias in neural networks

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50. In a perfect classification model, what would the AUC value be?
a) 0.5
b) 1
c) 0.75
d) 0.25

51. What is the Perceptron primarily used for?
a) Regression
b) Supervised learning for classification
c) Clustering
d) Dimensionality reduction

52. Which type of neural network contains multiple hidden layers?
a) Single-layer perceptron
b) Multilayer feedforward network
c) k-Nearest Neighbors
d) Decision Tree

53. Which of the following is NOT an activation function?
a) Sigmoid
b) ReLU
c) Tanh
d) Gradient Descent

54. What is the range of the Sigmoid activation function?
a) 0 to 1
b) -1 to 1
c) -∞ to ∞
d) 0 to ∞

55. Which activation function is commonly used in deep learning due to its non-linearity and efficiency?
a) Sigmoid
b) ReLU
c) Linear
d) Softmax

56. Which activation function is best suited for binary classification?
a) ReLU
b) Softmax
c) Sigmoid
d) Tanh

57. Which of the following activation functions can output negative values?
a) ReLU
b) Sigmoid
c) Tanh
d) Softmax

58. Which algorithm is used for training a neural network?
a) Decision Tree Algorithm
b) Backpropagation Algorithm
c) k-Means Algorithm
d) Random Forest Algorithm

59. What is the purpose of the Backpropagation algorithm?
a) To initialize weights randomly
b) To adjust weights using gradient descent based on errors
c) To increase the dataset size
d) To select the best features automatically

60. Which of the following is NOT a component of a neural network?
a) Weights
b) Biases
c) Hidden layers
d) Clustering centroid

MODULE 4

61. What does Maximum Likelihood Estimation (MLE) aim to maximize?
a) Posterior probability
b) Likelihood function
c) Prior probability
d) Mean squared error

62. Maximum a Posteriori (MAP) estimation incorporates which additional factor compared to MLE?
a) Likelihood function
b) Prior probability
c) Bias term
d) Gradient descent

63. What is the key difference between MLE and MAP?
a) MLE considers prior probability; MAP does not
b) MAP considers prior probability; MLE does not
c) MAP maximizes likelihood; MLE minimizes error
d) MLE is Bayesian; MAP is frequentist

64. Which decomposition explains the trade-off between bias and variance in model performance?
a) Principal Component Analysis
b) Singular Value Decomposition
c) Bias-Variance Decomposition
d) Fourier Decomposition

65. High bias in a model usually leads to:
a) Underfitting
b) Overfitting
c) Optimal generalization
d) No change in performance

66. Which statement best describes a model with high variance?
a) It captures noise along with patterns in training data
b) It generalizes well on new data
c) It underfits the data
d) It has low complexity

67. Which of the following techniques is used to reduce variance in models?
a) Increasing dataset size
b) Using more complex models
c) Removing regularization
d) Decreasing training data

68. If a model has high bias and low variance, it is likely to:
a) Perform well on the test set
b) Overfit the training data
c) Underfit the training data
d) Have high complexity

69. Which factor is NOT a cause of high bias in machine learning models?
a) Simple models
b) Insufficient features
c) Overly complex models
d) Strong regularization

70. Bias-variance decomposition applies mainly to which type of error?
a) Training error
b) Test error
c) Model complexity error
d) Random error

71. What is the main objective of a Support Vector Machine (SVM)?
a) Minimize classification error
b) Maximize margin between data classes
c) Minimize bias in the model
d) Increase model complexity

72. Which term is used to describe the boundary that separates classes in SVM?
a) Decision boundary
b) Hyperplane
c) Kernel
d) Loss function

73. What is the role of a soft margin in SVM?
a) It forces perfect classification
b) It allows misclassification for better generalization
c) It reduces the need for kernel functions
d) It removes non-linearity

74. Which technique is used in SVM to handle non-linearly separable data?
a) Principal Component Analysis
b) Kernel Trick

c) Logistic Regression
d) Naive Bayes

75. Which of the following is NOT a commonly used kernel in SVM?
a) Linear kernel
b) Polynomial kernel
c) Gaussian Mixture kernel
d) Radial Basis Function (RBF) kernel

76. What does the Radial Basis Function (RBF) kernel do in SVM?
a) Maps input data to a higher-dimensional space
b) Performs clustering
c) Works only for linear classification
d) Reduces computational complexity

77. Which parameter in SVM controls the trade-off between maximizing the margin and minimizing classification error?
a) Learning rate
b) C parameter
c) Number of hidden layers
d) Dropout rate

78. Which mathematical concept is fundamental to the Maximum Margin Classification in SVM?
a) Euclidean Distance
b) Lagrange Multipliers
c) Mean Squared Error
d) Cross-Entropy

79. What happens when an SVM model is trained with a very high C value?
a) It allows more misclassifications
b) It strictly minimizes classification errors but may overfit
c) It ignores the support vectors
d) It increases model bias

80. Which of the following is NOT true about SVM?
a) It can be used for both classification and regression
b) It performs well with high-dimensional data
c) It requires a large dataset to work efficiently

d) It is sensitive to the choice of kernel function

MODULE 5

81.Which of the following is NOT an ensemble learning method?
a) Voting
b) Bagging
c) Boosting
d) Backpropagation

82.What is the primary goal of ensemble learning?
a) To increase model complexity
b) To reduce variance and bias
c) To use a single classifier
d) To minimize dataset size

83. Which ensemble method creates multiple subsets of the training data and trains multiple models in parallel?
a) Boosting
b) Bagging
c) Stacking
d) Voting

84. Which of the following is an example of a bagging algorithm?
a) AdaBoost
b) Gradient Boosting
c) Random Forest
d) XGBoost

85. In Boosting, weak learners are trained in:
a) Parallel
b) Sequential manner
c) Random order
d) None of the above

86. Which of the following is a popular boosting algorithm?
a) Random Forest
b) AdaBoost
c) K-Means
d) KNN

87. What is the main disadvantage of Boosting?
a) It increases bias
b) It is prone to overfitting
c) It does not work with weak learners
d) It is less accurate than a single model

88. Which method aggregates predictions from multiple models using majority voting?
a) Bagging
b) Boosting
c) Stacking
d) Voting

89. Which ensemble method is most effective when models are highly diverse?
a) Bagging
b) Voting
c) Boosting
d) All of the above

90. Which of the following is true about Bagging?
a) It increases variance
b) It reduces overfitting
c) It does not require multiple models
d) It is less accurate than a single model

91. Which of the following is an unsupervised learning method?
a) Decision Trees
b) K-Means Clustering
c) Support Vector Machine
d) Logistic Regression

92. What is the main goal of clustering in machine learning?
a) Predict future data points
b) Find patterns and structures in data
c) Classify labeled data
d) Optimize loss functions

93. Which clustering algorithm divides data into 'k' groups based on similarity?
a) Hierarchical Clustering

b) K-Means
c) DBSCAN
d) PCA

94. In K-Means clustering, how are initial centroids selected?
a) Randomly
b) Based on prior knowledge
c) Using gradient descent
d) Using decision trees

95. Which clustering method is best for handling outliers?
a) K-Means
b) Hierarchical Clustering
c) DBSCAN
d) Mean-Shift

96. Expectation-Maximization (EM) is primarily used for which type of clustering?
a) Hard Clustering
b) Soft Clustering
c) Hierarchical Clustering
d) Density-Based Clustering

97. What does DBSCAN stand for?
a) Density-Based Spatial Clustering of Applications with Noise
b) Distance-Based Statistical Clustering Algorithm for Networks
c) Data-Based Systematic Clustering with Adaptive Normalization
d) Density-Bound Systematic Clustering of Adaptive Neurons

98. Which clustering method creates a dendrogram to represent data hierarchy?
a) K-Means
b) Hierarchical Clustering
c) DBSCAN
d) Expectation-Maximization

99. Which similarity measure is commonly used in clustering?
a) Euclidean Distance
b) Manhattan Distance
c) Cosine Similarity
d) All of the above

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100. Which clustering method allows clusters to have different shapes and sizes?
a) K-Means
b) DBSCAN
c) Hierarchical Clustering
d) Expectation-Maximization

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ANSWER KEYS

Module 1	Module 2	Module 3	Module 4	Module 5
1. (c)	21.(b)	41.(a)	61.(b)	81.(d)
2. (a)	22.(c)	42.(a)	62.(b)	82.(b)
3. (c)	23.(c)	43.(c)	63.(b)	83.(b)
4. (c)	24.(a)	44.(b)	64.(c)	84.(c)
5. (a)	25.(c)	45.(b)	65.(a)	85.(b)
6. (a)	26.(a)	46.(b)	66.(a)	86.(b)
7. (b)	27.(a)	47.(c)	67.(a)	87.(b)
8. (b)	28.(b)	48.(b)	68.(c)	88.(d)
9. (b)	29.(b)	49.(b)	69.(c)	89.(b)
10.(a)	30.(a)	50.(b)	70.(b)	90.(b)
11.(c)	31.(b)	51.(b)	71.(b)	91.(b)
12.(b)	32.(a)	52.(b)	72.(b)	92.(d)
13.(b)	33.(b)	53.(d)	73.(d)	93.(b)
14.(d)	34.(c)	54.(a)	74.(b)	94.(a)
15.(c)	35.(b)	55.(b)	75.(a)	95.(c)
16.(b)	36.(a)	56.(c)	76.(c)	96.(b)
17.(b)	37.(a)	57.(c)	77.(b)	97.(a)
18.(a)	38.(b)	58.(b)	78.(b)	98.(b)
19.(d)	39.(a)	59.(b)	79.(b)	99.(d)
20.(b)	40.(b)	60.(d)	80.(c)	100.(b)

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